

10/07/20

238 Product of Array Except Self:-

* Brute force :- $T_c \rightarrow O(n)$, $Sc \rightarrow O(n)$

Ex $\rightarrow [1, 2, 3, 4]$

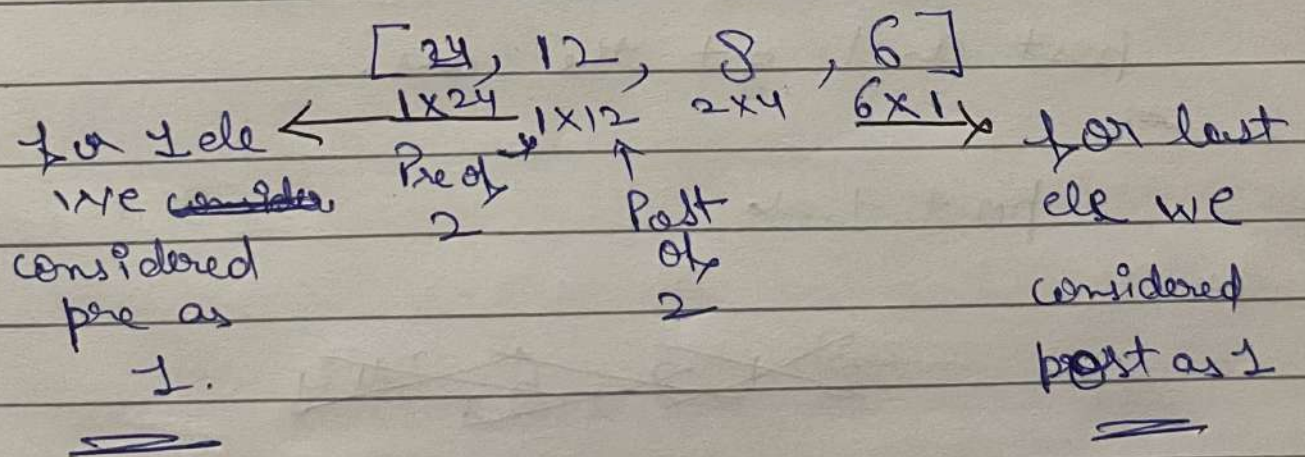
Pre $\rightarrow [1, 2, 6, 24]$

Post $\rightarrow [24, 24, 12, 4]$

$\rightarrow O(n)$

Problem without this we can't

Output $\rightarrow$ Pre of ele $\times$ Post of ele



* Optimize :- $T_c \rightarrow O(n)$, $Sc \rightarrow O(1)$

arr $\Rightarrow$ $\{1, 2, 3, 4\}$

$\Rightarrow$ Store Prefix product first in the arr.

output $\Rightarrow$ $\{1, 1, 2, 6\}$

~~output $\Rightarrow$ $\{1, 2, 6, 24\}$~~

Here we stored Prefix product into the arr.

$\Rightarrow$ After storing the Prefix product with the use of post as a variable we can also store the post and get the ans.

post $\Rightarrow$

~~$\{1, 2, 6, 24\}$~~

nums $\Rightarrow$ $\{1, 2, 3, 4\}$

~~output $\Rightarrow$ $\{1, 2, 6, 24\}$~~

~~post $\Rightarrow$ 1~~

ans(3) $\rightarrow$ post.

~~0~~

~~post $\rightarrow$ 6~~

post $\rightarrow$ 24, 12, 8, 6

24

$i \rightarrow 3$

i

out put $\rightarrow$ 1, 1, 2, 6

$i \rightarrow 2$

i

$\rightarrow$ 1, 1, 2, 6
8

$i \rightarrow 1$

i

$\rightarrow$ 1, 1, 8, 6
12

$i \rightarrow 0$

i

$\rightarrow$ 1, 12, 8, 6
24

After loop:-

ans $\rightarrow$ 24, 12, 8, 6